

# On the Invariant Theory of Forms in $n$ Variables

By Miss *Emmy Noether* in Erlangen.

*Journal für die reine und angewandte Mathematik* 139 (1911), pp. 118–154

*Source-audited working draft containing §4 only; original printed pp. 132–134.*

## § 4. The Decomposition Identities.

We now consider the most general case of identities between arbitrary symbol and variable rows in whose final expressions, without performing substitution (11), a row  $p_\tau$  occurs decomposed into its individual rows  $y^{(1)} \dots y^{(\tau)}$ ; we call these identities “decomposition identities.” In accordance with the remark at the end of the preceding section, we first determine them for the case of only two symbol rows, and thus expand the expression  $(q_\rho A^\sigma \mid B_{\rho+\sigma-\tau} p_\tau)$ . We set

$$q_{\rho-\alpha}^{(i_1 \dots i_\alpha)} \sim p_{n-\rho} y^{(i_1)} \dots y^{(i_\alpha)}, \quad p_{\tau-\alpha}^{(i_1 \dots i_\alpha)} = y^{(i_{\alpha+1})} \dots y^{(i_\tau)}, \quad p_\tau = y^{(1)} \dots y^{(\tau)}. \quad (22)$$

With respect to the numbers  $\rho, \sigma, \tau$ , four cases must be distinguished:

- 1)  $\rho + \sigma \leq n, \quad \tau \leq \rho, \quad \tau \leq \sigma;$
- 2)  $\rho + \sigma \leq n, \quad \tau \geq \rho, \quad \tau \leq \sigma;$
- 3)  $\rho + \sigma \leq n, \quad \tau \leq \rho, \quad \tau > \sigma;$
- 4)  $\rho + \sigma \leq n, \quad \tau \geq \rho, \quad \tau \geq \sigma,$

of which we shall single out the first and then briefly characterize the remaining ones. By (20), (10), and (9), if we decompose the row  $y^{(1)} y^{(2)} \dots y^{(\tau)} B_{\rho+\sigma-\tau}$  into  $y^{(1)}$  and  $y^{(2)} \dots y^{(\tau)} B_{\rho+\sigma-\tau}$ , we obtain

$$(q_\rho A^\sigma \mid y^{(1)} y^{(2)} \dots y^{(\tau)} B_{\rho+\sigma-\tau}) = (q_{\rho-1}^{(1)} A^\sigma \mid y^{(2)} \dots y^{(\tau)} B_{\rho+\sigma-\tau}) + (-1)^\rho (q_\rho [A_{n-\sigma} y^{(1)}]^\sigma \mid y^{(2)} \dots y^{(\tau)} B_{\rho+\sigma-\tau}). \quad (23)$$

The last term in (23) now has exactly the form of the original expression—only  $\sigma$  is replaced by  $\sigma - 1$ , and  $\tau$  by  $\tau - 1$ —and therefore again admits equation (23). Thus, since  $\tau \leq \sigma$ , we can apply operation (23)  $\tau$  times and, taking account of (10), arrive at the relation

$$(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) = (-1)^{\rho\sigma+(\sigma+\tau)(n+1)} (q_\rho B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_\tau) + \sum_{i_1=1}^{\tau} (-1)^{\rho(i_1-1)} (q_{\rho-1}^{(i_1)} [A_{n-\sigma} y^{(1)} \dots y^{(i_1-1)}]^\sigma \mid y^{(i_1+1)} \dots y^{(\tau)} B_{\rho+\sigma-\tau}). \quad (24)$$

Every term under the summation sign is again of the type of the original expression:  $\rho$  is replaced by  $\rho - 1$ ,  $\sigma$  by  $\sigma - i_1 + 1$ , and  $\tau$  by  $\tau - i_1$ , so condition 1) is satisfied even more strongly. We can therefore, since  $\tau \leq \rho$ , apply operation (24)  $\tau$  times and obtain the desired decomposition identity:

$$\begin{aligned} (q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) &= \varepsilon_{i_0} (q_\rho B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_\tau) \\ &+ \sum_{i_1=1}^{\tau} \varepsilon_{i_1} (q_{\rho-1}^{(i_1)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-1}^{(i_1)}) \\ &+ \sum_{i_2=i_1+1}^{\tau} \varepsilon_{i_2} (q_{\rho-2}^{(i_1 i_2)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-2}^{(i_1 i_2)}) + \dots \\ &+ \varepsilon_{i_\tau} (q_{\rho-\tau}^{(i_1 \dots i_\tau)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma}) \\ &= \sum_{\alpha=0}^{\tau} \sum_{i_\alpha=i_{\alpha-1}+1}^{\tau} \varepsilon_{i_\alpha} (q_{\rho-\alpha}^{(i_1 \dots i_\alpha)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-\alpha}^{(i_1 \dots i_\alpha)}). \end{aligned} \quad (25)$$

For the sign  $\varepsilon$ , (24) gives

$$\begin{aligned} \varepsilon_{i_\alpha} &= (-1)^{\delta_{i_\alpha}}, \quad \delta_{i_\alpha} = \sum_{\beta=1}^{\alpha} (\rho - \beta + 1)(i_{\beta-1} + i_\beta + 1) \\ &+ (\rho - \alpha)(\sigma + i_\alpha + \alpha) + (\sigma + \tau + \alpha)(n + 1), \end{aligned} \quad (26)$$

where  $i_0 = 0$  is to be set. The summation sign in (25) is to be understood as follows: to every combination  $i_1 i_2 \dots i_{\alpha-1}$ , where  $i_1 < i_2 < \dots < i_{\alpha-1}$ , all values  $i_\alpha$  for which  $i_{\alpha-1} < i_\alpha \leq \tau$  are adjoined. As one readily verifies on taking (26) into account, the individual sums in (25) satisfy the differential equations analogous to (19):

$$\left( \frac{\partial}{\partial y^{(i)}} \middle| y^{(i+1)} \right) = 0, \quad (i = 1, 2, \dots, \tau - 1). \quad (27)$$

They therefore depend only on the row  $p_\tau$ . Thus the decomposition identity is an indecomposable identity between the rows  $A, B, q, p$ ; we shall return to its explicit representation in the next section.

In case 2), operation (23) can likewise be applied  $\tau$  times, but operation (24) stops after the  $\rho$ -th application. Thus, in the final expression of (25), the first summation sign is to be replaced by  $\sum_{\alpha=0}^\rho$ . In cases 3) and 4), operation (23) can be carried out only  $\sigma$  times; in place of (24) one then has the relation<sup>1</sup>

$$\begin{aligned} (q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) &= (-1)^{\rho\sigma} (q_{\rho-(\tau-\sigma)}^{(\sigma+1\dots\tau)} B^{n-\rho+\tau-\sigma} \mid A_{n-\sigma} p_{\tau-(\tau-\sigma)}^{(\sigma+1\dots\tau)}) \\ &+ \sum_{i_1=1}^\sigma (-1)^{\rho(i_1-1)} (q_{\rho-1}^{(i_1)} [A_{n-\sigma} y^{(1)} \dots y^{(i_1-1)}]^\sigma \mid y^{(i_1+1)} \dots y^{(\tau)} B_{\rho+\sigma-\tau}); \end{aligned} \quad (28)$$

repeating (28)  $\tau - \sigma$  times, with the summation sign changing to  $\sum_{i_\beta=i_{\beta-1}+1}^{\sigma+\beta-1}$ —as is shown by applying (23)  $\sigma - i_1 + 1$  times to the terms under the summation sign in (28)—gives all the terms of the individual sum in (25) corresponding to the index  $\alpha = \tau - \sigma$ , with their respective signs. Thereafter cases 3) and 4) pass into cases 1) and 2), since the numbers corresponding to  $\sigma, \tau$  become  $\sigma' = \sigma - i_{\tau-\sigma} + \tau - \sigma$ ,  $\tau' = \tau - i_{\tau-\sigma} = \sigma'$ ; and as the procedure continues,  $\tau' < \sigma'$ . Thus, in the final expression of (25), the first summation sign is to be replaced by  $\sum_{\alpha=\tau-\sigma}^{\tau,\rho}$ .

Analogously, (21) gives the dual decomposition identity, in which a row  $q_\rho$  is decomposed into the individual rows  $v^{(1)}, \dots, v^{(\rho)}$ . If we set

$$\dot{p}_{\tau-\beta}^{(i_1\dots i_\beta)} \sim v^{(i_1)} \dots v^{(i_\beta)} q_{n-\tau}; \quad \dot{q}_{\rho-\beta}^{(i_1\dots i_\beta)} = v^{(i_{\beta+1})} \dots v^{(i_\rho)}; \quad q_\rho = v^{(1)} \dots v^{(\rho)}, \quad (29)$$

then

$$(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) = \sum_{0, \tau-\sigma}^{\tau, \rho} \sum_{i_\beta=i_{\beta-1}+1}^\rho \varepsilon(\dot{q}_{\rho-\beta}^{(i_1\dots i_\beta)} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} \dot{p}_{\tau-\beta}^{(i_1\dots i_\beta)}). \quad (30)$$

Here the summation index  $\beta$  runs from the larger of the lower values to the smaller of the upper values. All terms of an individual sum in (25) and (30) are polars of a single form, obtained through the polar processes

$$\left( \frac{\partial}{\partial p_{\tau-\alpha}} \middle| p_{\tau-\alpha}^{(i_1\dots i_\alpha)} \right), \quad \left( q_{\rho-\alpha}^{(i_1\dots i_\alpha)} \middle| \frac{\partial}{\partial q_{\rho-\alpha}} \right).$$

To express this, we denote by  $\Pi$  an aggregate of such polar operations, multiplied by constants, and obtain, in place of (25) and (30), the relation

$$(q_\rho A^\sigma \mid p_\tau B_{\rho+\sigma-\tau}) = \sum_{0, \tau-\sigma}^{\tau, \rho} \Pi(q_{\rho-\alpha} B^{n-\rho-\sigma+\tau} \mid A_{n-\sigma} p_{\tau-\alpha}). \quad (31)$$

---

<sup>1</sup>In the first term of (28), the original print has  $q_{\rho-(\tau-\sigma)}^{(\sigma+1\dots\tau)}$  and  $p_{\tau-(\tau-\sigma)}^{(\sigma+1\dots\tau)}$ . R823 and the inherited English witness instead have  $q_\rho^{(\sigma+1\dots\tau)}$  and  $p_{\tau-\sigma}^{(\sigma+1\dots\tau)}$ ; the print is followed here.